

Chapter 3: Elasticity of Demand and Supply

Elasticity of Demand and Supply

1. Elasticity of Demand

Elasticity of demand measures how much the **quantity demanded** of a commodity change in response to changes in its **determinants** (price, income, or price of related goods).

Types of Elasticity of Demand

1. Price Elasticity of Demand
 2. Income Elasticity of Demand
 3. Cross Elasticity of Demand
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A. Price Elasticity of Demand (E_p)

It shows the **responsiveness of demand** to a change in the **price** of a commodity.

$$E_p = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

where,

E_p = Coefficient of price elasticity of demand

Q = Initial quantity demanded

ΔQ = Change in Quantity demanded

ΔP = Change in Price

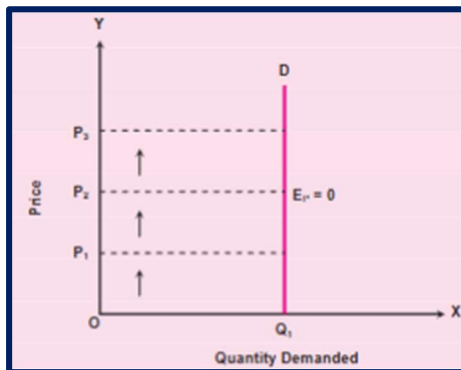
P = Initial Price

Types of Price Elasticity:

1. **Perfectly Elastic Demand ($E_p = \infty$):** Demand changes infinitely even with a slight change in price.



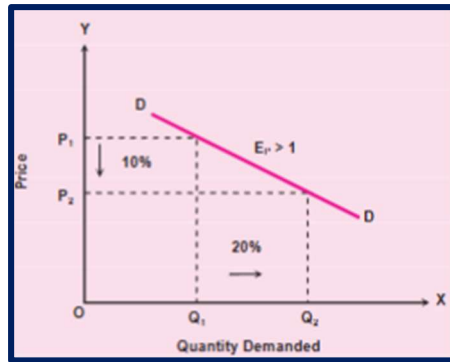
2. **Perfectly Inelastic Demand ($E_p = 0$):** Demand remains constant despite price change.



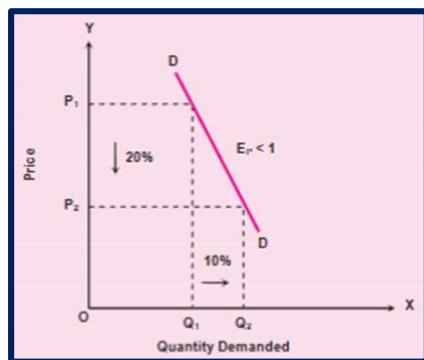
3. **Unitary Elastic Demand ($E_p = 1$):** % change in demand equals % change in price.



4. **Relatively Elastic Demand ($E_p > 1$):** Demand changes more than price.



5. **Relatively Inelastic Demand (E_p < 1)**: Demand changes less than price.



B. Income Elasticity of Demand (E_y)

It measures the **responsiveness of demand** to a change in the **income** of the consumer.

$$E_Y = \frac{\Delta Q}{\Delta Y} \times \frac{Y}{Q}$$

where,

E_Y = Coefficient of income elasticity of demand

Q = Initial quantity demanded

ΔQ = Change in Quantity demanded

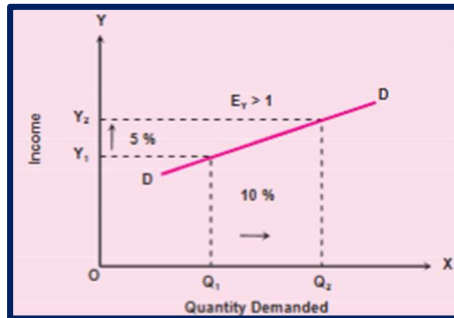
ΔY = Change in Income

Y = Initial Income

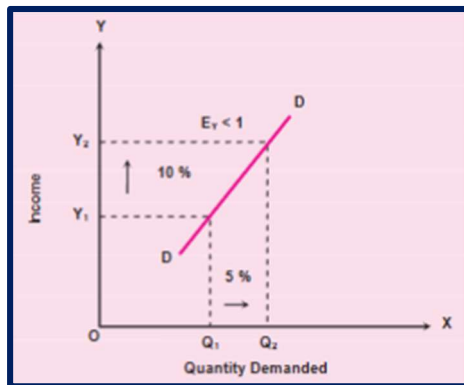
Types of Income Elasticity:

1. **Positive (E_y > 0)**: Demand rises with income.

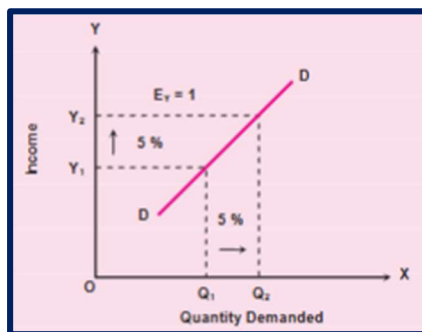
- Greater than unity ($E_y > 1$) → The percentage increase in demand is more than percentage increase income and vice versa. *For Ex. Luxury goods*



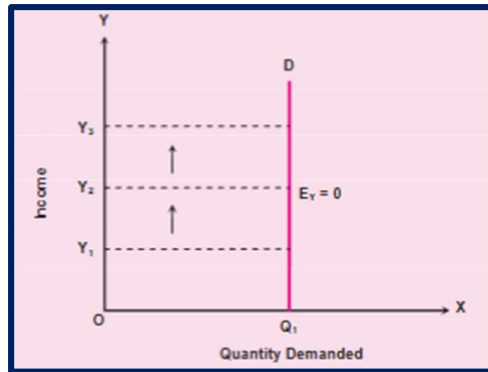
- Less than unity ($E_y < 1$) → The percentage increase in demand is less than percentage increase income and vice versa. *For Ex. Necessities*



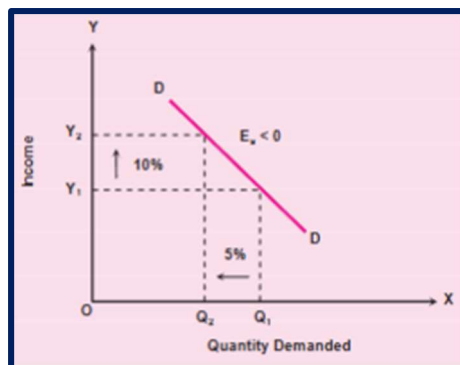
- Equal to unity ($E_y = 1$) → The percentage increase in demand is Equal to percentage increase income and vice versa. *For Ex. Normal goods*



2. **Zero ($E_y = 0$):** Demand unaffected by income (e.g., salt).



3. **Negative ($E_Y < 0$):** Demand falls as income rises (inferior goods).



C. Cross Elasticity of Demand (E_{XY})

It shows how the **demand for one good (X)** responds to the **price change of another good (Y)**.

$$E_{XY} = \frac{\Delta Q_X}{\Delta P_Y} \times \frac{P_Y}{Q_X}$$

where,

E_{XY} = Coefficient of Cross elasticity of demand

ΔQ_X = Change in the demand for commodity-X

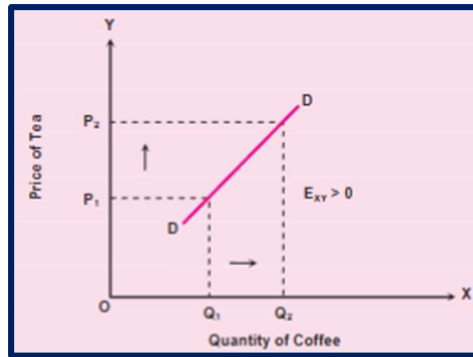
ΔP_Y = Change in price of commodity-Y

P_Y = Price of commodity-Y

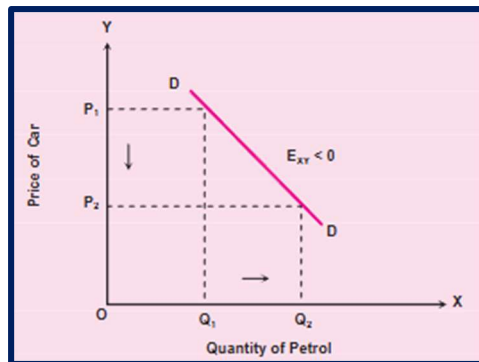
Q_X = Quantity of commodity-X

Types of Cross Elasticity:

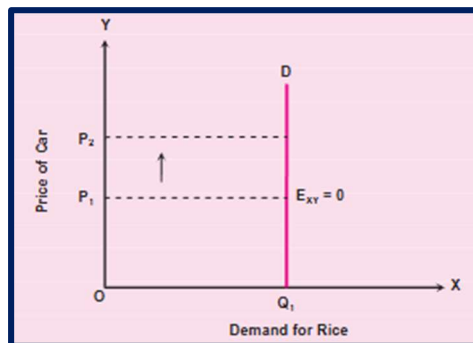
1. **Positive ($E_{XY} > 0$):** Substitute goods (tea and coffee).



2. **Negative ($E_{xy} < 0$):** Complementary goods (car and petrol).



3. **Zero ($E_{xy} = 0$):** Unrelated goods.



Determinants of Elasticity of Demand

1. Nature of the commodity
2. Availability of substitutes
3. Number of uses
4. Income level of consumers
5. Joint demand

6. Postponement of consumption
 7. Habits
 8. Price level
 9. Time period
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2. Elasticity of Supply

Elasticity of supply measures how much the **quantity supplied** of a commodity change with a change in its **price**.

$$E_s = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

where,

E_s = Coefficient of elasticity of supply or price elasticity of supply

P = Initial price

Q = Initial quantity supplied

ΔQ = Change in Quantity Supplied

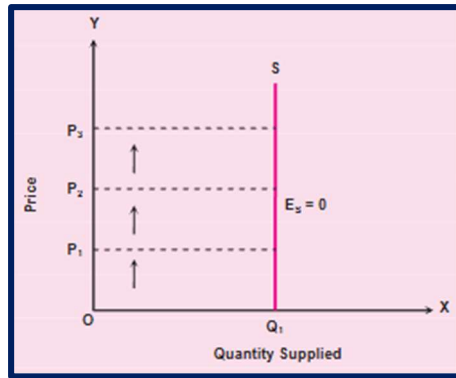
ΔP = Change in price

Types of Elasticity of Supply:

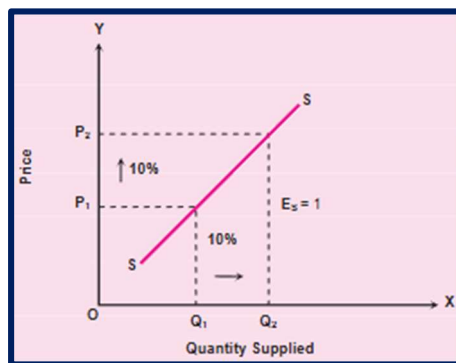
1. **Perfectly Elastic ($E_s = \infty$):** If negligible change in the price results in the infinite or unlimited change in the quantity supplied.



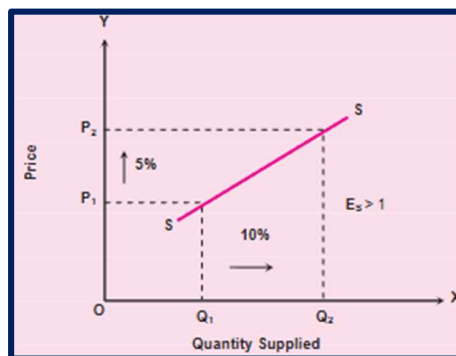
2. **Perfectly Inelastic ($E_s = 0$):** If there is no change in the quantity supplied despite the change in the price of the commodity.



3. **Unitary Elastic ($E_s = 1$):** If the percentage change in quantity supplied is equal to the percentage change in the price of the commodity.



4. **Relatively Elastic ($E_s > 1$):** If the percentage change in the quantity supplied of a commodity is more than the percentage change in the price of the commodity.



5. **Relatively Inelastic ($E_s < 1$):** If the percentage change in the quantity supplied of a commodity is less than the percentage change in the price of the commodity.

